

CPS209 Lab 5

Comparing Polynomials

Preamble

In this lab you will continue to build on your Polynomial class by overriding the equals() method, implementing the Comparable interface, and adding another arithmetic operation - multiplication.

Homework to Complete (Submit on D2L):

This lab requires you to continue working in Polynomial.java (the same one from previous labs) An updated tester for Polynomial.java is given.

Reminders:

- Your methods should be SILENT. Print statements are fine for testing and debugging but remove them from your code before you submit.
- No method should take longer than a couple of seconds to pass the tester. If your code takes minutes (or *hours*, heaven help you) it means you are doing something very inefficient in your approach and/or implementation. Remedy this before submitting. Our TAs will be very busy and cannot wait 20 minutes for a single student's code to execute.

Method Descriptions – Polynomial.java:

public Polynomial multiply(Polynomial other)

Creates and returns a brand-new Polynomial object that represents the result of polynomial multiplication of this and other. Multiplication of two polynomials can cancel out some internal terms whose coefficients were originally nonzero in both polynomials. For example, multiplying $x^2 + 3$ by $x^2 - 3$ gives the result $x^4 - 9$ whose second order term ends up with a zero coefficient and vanishes from the result. However, unlike when adding two polynomials, the highest term can never get cancelled out this way, since the product of two nonzero integer coefficients can never be zero. You therefore know the degree of the entire result of multiplication right away once you know the degrees of the original polynomials, without actually having to carry out this multiplication by adding all the terms together.

@Override public boolean equals(Object other)

Returns true if the other object is also a Polynomial of the exact same degree as this, and that the coefficients of this and other polynomials are pairwise equal. If the other object is anything else, this method should return false.

public int compareTo(Polynomial other)

Implements the **ordering comparison** between this and other polynomials, as required by the interface Comparable<Polynomial>. This method should return +1 if **this** is greater than the **other**, -1 if the **other** is greater than **this**, and return a 0 if both polynomials are equal in the sense of the equals method.

A total ordering relation between polynomials can be defined by many possible different rules. We shall use an ordering rule that says that **any polynomial of a higher degree is automatically greater than any polynomial of a lower degree**, regardless of their coefficients. For two polynomials whose degrees are equal, the result of the order comparison is determined by **the highest-order term for which the coefficients of the polynomials differ**. For example, consider the two polynomials below:

i. $x^2 + 3x + 12$

ii. $x^2 + 5x + 2$

Polynomial ii is greater, because it has a larger coefficient on the highest order term whose coefficients differ ($5 > 3$).

Submission

Labs are to be completed and submitted *individually*. Submit your completed **Polynomial.java** on D2L, under the submission for Lab 5.